Soheil Zibakhsh Shabgahi

Computer Engineering · PhD Candidate at UCSD

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Research Interests

Agentic AI & Reinforcement Learning; LLM Training, Post-Training & Inference-Time Scaling; Efficient Model Architectures & ML Optimization.

Related Skills

Research Areas	Large Language Models, Mixture-of-Experts (MoE), Reinforcement Learning, Agentic AI, Inference-Time Scaling, Model Compression, Adaptive Inference, Model Optimization.
ML Frameworks & Systems	PyTorch, JAX, Hugging Face Transformers, vLLM, Triton, Distributed Computing, GPU Computing, Multithreading.
Programming & Tools	Python, C/C++, MATLAB, Verilog, Git, Linux, LaTeX.

Research & Work Experience

ACES Lab UCSD — *University of California San Diego*

Graduate Student Researcher — *Sep 2022 - PRESENT*

- Conducting research with Professor Farinaz Koushanfar at the ACES lab in the Machine Intelligence and Computer Security (MICS) Center. Research LLM training and inference, reinforcement learning, model compression, and optimization, focusing on model capability, efficiency, and robustness.
- Managing lab server affairs, including maintenance and resource management for 12 GPU servers and a NAS server.

Agentic AI Researcher and Chipsage Labs — *Menlo park*

Researcher — *August 2026 - September 2026*

- Researched graph-based memory architectures and retrieval mechanisms for LLM-based agents in chip design workflows.
- Developed agent memory mechanisms supporting quality assessment, test-time learning, and feedback-driven improvement.

Data and ML Innovation (DMLI) at Apple — *Apple*

ML Researcher — *March 2025 - September 2025*

- Conducted research on Mixture-of-Experts (MoE) architectures, focusing on inference-time scaling, adaptive expert selection, and accuracy–compute tradeoffs.
- Developed RoE, a training-free inference algorithm that enables a 7B MoE model to match the prediction performance of a 10.5B model using 30% less inference compute.
- Contributed to MoE-PHDS, enabling flexible runtime sparsity and configurable accuracy–latency tradeoffs using a single model checkpoint.
- Research resulted in two papers on efficient LLM inference and MoE optimization.

University of Tehran Data Lab — *University of Tehran, ECE*

Student Researcher — *Jan. 2021 - Apr. 2022*

- Conducting a research on the evaluation of Bitcoin Lightning Networks.
- The research was done under the supervision of Dr.B.Bahrak and Dr.P.Shariatpanahi

Education

University of California San Diego — *La Jolla, USA*

PhD. in Computer engineering — *2022 - Exp. 2027 (06/27)*

- ACES Lab under supervision of Professor Farinaz Koushanfar.
- Thesis: Scalable and Adaptive Training Frameworks for Reinforcement Learning and Language Models
- Expected to graduate June 2027 (06/27)

University of California San Diego — *La Jolla, USA*

MS. in Computer engineering — *2022 - 2024*

University of Tehran — *Tehran, Iran*

B.Sc. in Computer engineering — *2018 - 2022*

- GPA : 4/4

Teaching

Optimization and Acceleration of Deep Learning on Various Hardware Platforms *UCSD*

Teaching Assistant *Winter&Fall 2026*

- Served as Teaching Assistant for two offerings of a graduate-level course on hardware acceleration for machine learning.
- Developed instructional materials connecting machine learning algorithms with hardware-aware implementation and computational efficiency.

Honors & Awards

2026 **Qualcomm Innovation Fellowship**, Finalist.
2022 **Fellowship**, PhD student at University of California San Diego.
2018-2021 **Ranked first at the University of Tehran**, among all Computer Engineering students.
2017 **Silver medal**, in the 30th Iranian National Olympiad in physics.
2018 **Ranked**, Top 0.6% national-wide university entrance exam
2017 **Becoming a member**, of Iran's National Elites Foundation(NODET)
2018 **Scholarship**, Received from the University of Tehran Sponsors Foundation as an exceptional talent student.

Selected Publications

MoEs Are Stronger than You Think: Hyper-Parallel Inference Scaling with RoE *Findings of EMNLP 2026*

S.Zibakhsh, M.Samragh, K.Nishu, L.Hannah, A.Kundu and M.Cho. Pre-print at Link *2026*

Introduced Roster of Experts (RoE), a training-free MoE inference-time scaling algorithm enabling a 7B model to match a 10.5B model's prediction quality with 30% less inference compute.

ForTIFAI: Fending Off Recursive Training Induced Failure for AI Models *Nature Partner Journal Artificial Intelligence*

S.Zibakhsh*, P.Aghazadeh*, F.Koushanfar and A.Mirhoseini. link at Link *2026*

Developed Truncated Cross-Entropy (TCE), a novel loss function mitigating synthetic-data-induced model collapse, enabling models to tolerate 2.3× more synthetic data before performance degradation; introduced an open-source model-collapse benchmark.

Beyond Perplexity: A Lightweight Benchmark for Knowledge Retention in Supervised Fine-Tuning *Under Review*

S.Zibakhsh*, P.Aghazadeh*, F.Koushanfar . Pre-print at Link *2026*

Introduced the Knowledge Retention (KR) Test, a lightweight, corpus-grounded benchmark for evaluating factual knowledge acquisition and retention during LLM supervised fine-tuning, beyond conventional perplexity metrics.

MoE-PHDS: One MoE checkpoint for flexible runtime sparsity

L.Hannah, **S.Zibakhsh**, K.Nishu, A.Kundu, M.Samragh, M.Farajtabar and M.Cho. Link *2025*

Helped with conceptualizing and implementing a lightweight MoE fine-tuning approach enabling dynamic inference-time sparsity and configurable accuracy–latency tradeoffs from a single checkpoint, matching or exceeding models optimized for individual deployment targets.

LayerCollapse: Adaptive compression of neural networks *Under Review*

S.Zibakhsh, M.Shariff and F.Koushanfar. Pre-print at Link *2023*

Developed LayerCollapse, an adaptive model-compression method using dynamic regularization to optimize neural network depth for different tasks, achieving state-of-the-art results across multiple applications.

Additional Publications

MergeGuard: Efficient Thwarting of Trojan Attacks in Machine Learning Models

S.Zibakhsh*, Y.Jandali* and F.Koushanfar. Pre-print at Link *2023*

Developed MergeGuard, a post-training layer-merging method that improves model generalization and mitigates trojan attacks while preserving Transformer accuracy, outperforming conventional fine-tuning defenses.

LiveTune: Dynamic Parameter Tuning for Feedback-Driven Optimization *Under Review*

S.Zibakhsh, N.Shaybani, A.Tabrizi and F.Koushanfar. Pre-print at Link *2023*

Developed LiveTune, a Python framework for real-time hyperparameter adjustment during model training, eliminating training restarts and improving training efficiency and resource utilization.

*Equal Contribution.

Closing the Loop: Overcoming the Tool-Feedback Gap in Agentic Hardware Design *ISQED 2026*

M.Tarighat, K.I.Gubbi, N.Javidnia, N.Marasok, **S.Zibakhsh**, M.Halem, M.Pirayesh, S. Rafatirad, F.Koushanfar, H.Homayoun. paper at IEEE *2026*

Co-authored a survey of closed-loop LLM agents for hardware design, examining EDA tool feedback, multi-agent orchestration, and challenges in iterative RTL generation and verification.

Agentic AI for Chip Design Verification: Failure Modes, Metrics, and Coverage Closure *ISQED 2026*

N.Marasok, M.Halem, K.I.Gubbi, M.Tarighat, N.Javidnia, **S.Zibakhsh**, K.Huang, S. Rafatirad, H.Sayadi, F.Koushanfar, H.Homayoun. paper at IEEE *2026*

Co-authored a study of agentic chip verification, examining how autonomous reasoning and tool execution can bridge high-level design intent and silicon-accurate verification.

Modeling Effective Lifespan of Payment Channels

S.Zibakhsh, M.Hosseini, B.Bahrak and S.Shariatpanahi. Preprint at Link *2022*

Developed a random-walk model and theoretical analysis to characterize the effective lifespan of Bitcoin Lightning Network payment channels before recharging.

Throughput limitation of the off-chain payment networks

S.Dehshali*, S.Hosseini*, **S.Zibakhsh***, B.Bahrak and S.Shariatpanahi. Preprint at ePrint *2022*

Analyzed the graph structure of Bitcoin Lightning Networks to characterize throughput limitations in off-chain payment systems.